Lab 2 – Light or Paint? Intrinsic Images

CV8508 – Perceptual Cognition for Computer Vision

An image is the product of two things: the *reflectance* R of the surfaces (paint) and the *shading* S of the light, $I = R \cdot S$. Lecture 4 showed that people recover R from I using cues such as penumbra, ratios across edges and co-planarity. You will build two decompositions, one from a physical prior and one learned from human judgments in Intrinsic Images in the Wild [1], test both on scenes built to confuse them, and then ask a recent multimodal model to do the same. About 60 minutes; a coding agent is allowed, but check what it does.

Setup

```
mkdir lab2 && cd lab2
wget https://utkarshmall13.github.io/cv8508-labs/lab2/setup.sh
sh setup.sh                       # downloads data and lab2_tools.py, builds .venv
source .venv/bin/activate
```

Read `lab2_tools.py` first (loaders, the WHDR metric, probe measurements; no decomposition code). `data/` then contains:

`iiw` a downsized IIW subset, `train/` and `test/`. For each photo, humans compared pairs of points: "same reflectance", "left is darker" or "right is darker", with a confidence weight. `load_iiw()` returns the images and judgments. *Never train on `test`.*

`confusing` six synthetic scenes with known albedo, `scenes.json` with *probe* boxes and the true albedo ratio of probe pairs. Pixel values of the two Adelson-style probes are identical.

Tasks

Save every step as a script. Work in log space and in linear light (convert from sRGB first).

1. **Method A: physics and a simple prior, no training data (15 min).** Decompose each image into R and S using a prior you can state in one sentence for each (for example shading is smooth and reflectance is piecewise constant, the Retinex idea [2]). Report the mean WHDR on `test`. Check that your solver has converged; smooth fields converge slowly.

2. **Method B: learned from human judgments (20 min).** Train a small CNN on `train` that maps an image to log reflectance, supervised only by the human judgments (plus any prior you want). Keep it under about 500 000 parameters, train on the 160 px images for no more than 6 epochs, and expect roughly 10 minutes on a laptop CPU. Report WHDR on `test`, next to Method A and next to the trivial answer "everything has the same reflectance". *Hint:* about two thirds of the judgments say "equal", so a net that outputs log reflectance directly tends to collapse to a constant. Try predicting log shading and setting $\log R = \log I - s$ (start from $s = 0$).

3. **Confusing scenes (10 min).** Run both methods on all six scenes. Make one contact sheet (image, truth, A, B) and one table of probe ratios with `probe_ratio`: pixels, truth, A, B.

4. **A recent multimodal model (10 min).** Give the six images to a recent multimodal model that can return images (for example the Gemini app) and ask for an albedo image and a shading image. Also ask in words which probe is painted darker. Save what it returns,

convert to linear light, and measure the same probe ratios. Record your prompts. Then repeat the three methods on one photo of your own that has a shadow on a floor or wall.

Use a coding agent for the plumbing

Agents are welcome for writing plotting code, running models and tidying figures. Check what they produce: open the images, test on one small input, and make sure you can explain every step. The *inference* — what the results show, why, and what surprised you — must be written in your own words.

Optional

Try Adelson's original checker-shadow image [3]. Or add one new confusing scene of your own with known R and S where you expect A and B to disagree.

Inference – write this part yourself

Answer in a few sentences each, citing your own numbers, and mark each claim as evidence, hypothesis or speculation.

1. How do A and B compare on WHDR, and how do both compare with the trivial answer? With about 300 test images, how large a gap would be meaningful?
2. On the soft and the hard checker shadow, what ratio did each method recover (truth 3.75, pixels 1.0)? Which cue from the lecture does each one use, and which is missing?
3. Which scene separates A and B most? Explain it from what each method assumes or has learned. Which scenes fool a method in the same way they fool people?
4. Where is a method closer to the truth than a person would be?
5. Did the multimodal model's words and its picture agree? Where did it fail, and could you tell without the ground truth?
6. Propose one experiment that would show whether a model uses shadow cues or just brightness context.

References

[1] Sean Bell, Kavita Bala, and Noah Snavely. Intrinsic images in the wild. *ACM Transactions on Graphics (SIGGRAPH)*, 33(4), 2014.

[2] Edwin H. Land and John J. McCann. Lightness and retinex theory. *Journal of the Optical Society of America*, 61(1):1–11, 1971.

[3] Edward H. Adelson. Checker shadow illusion. https://persci.mit.edu/gallery/checkershadow.